

Design Evaluation 1—XR Podcast Explorer

1. Objective and Validation Metrics

The objective of this evaluation was to determine whether first-time users could understand and complete the main spatial interaction flow of the **XR Podcast Explorer** without detailed instructions. Five participants completed a moderated task-based usability test covering the journey from **Podcast Hub** → **Episode Room** → **Chapter Portal** → **Podcast Experience**, followed by Favourite and return-navigation tasks.

Success was evaluated through **task completion, hesitation, errors, assistance required, direct observation and think-aloud feedback**. An interaction was considered successful when participants could understand its purpose and complete it independently with minimal hesitation. Repeated hesitation or moderator assistance was treated as evidence that further iteration was required.

Interaction tested	Validation criterion
Topic Door	Recognise the door as an entry point and enter the AI topic
Episode	Browse available Episodes and select one
Chapter	Browse Chapters, select Interview and enter it
Favourite	Understand and complete the Heart interaction independently
Navigation	Return to Episode Room and Podcast Hub
Spatial flow	Understand Topic → Episode → Chapter → Experience hierarchy

2. Results

Testing showed that the **core podcast exploration flow was generally understandable**, but interaction performance varied. Door-based entry and Episode browsing were the strongest interactions. The Favourite interaction produced the most repeated difficulty, while Chapter transitions and global navigation showed some uncertainty.

Area	Result	Key finding
Topic Door	Strong	Doors were generally understood as spatial entry points and were positively received.
Episode Browsing	Strong	Participants could browse and select Episodes; button-based controls were considered clear.
Chapter Selection	Mixed	Tasks were completed, but Chapter meaning and the Chapter → Current Chapter Door transition caused uncertainty.
Favourite	Weak	Multiple participants needed guidance to manipulate the Heart, despite generally recognising its favourite meaning.
Navigation	Mixed	Return navigation was understood overall, but panel behaviour and spatial relationships caused uncertainty.
XR Experience	Mixed	Doors supported immersion, but participants also described the environment as static or insufficiently spatial.

Recurring feedback included requests for clearer interaction cues, a more physical Heart, stronger spatial relationships between rooms, richer environmental content and improvements to supporting UI.

3. Analysis / Insights

Insight 1 — Spatial doors are the strongest part of the XR interaction

Doors consistently communicated the idea of **entering podcast content**. Participants could connect the familiar physical action of entering a door with navigating between digital content spaces.

This validates the use of a spatial metaphor rather than relying entirely on conventional menus. **Topic Door and Current Chapter Door interactions should therefore be retained and strengthened.**

Insight 2 — The Heart metaphor works, but its affordance does not

Participants often understood that the Heart represented **favouriting or collecting content**, yet several did not understand that they needed to physically move it to the Favourite Area.

This reveals a distinction between **semantic clarity and interaction clarity**: the concept is understandable, but the required action is not sufficiently discoverable. The Heart metaphor should therefore remain, while its size, three-dimensionality, grab feedback and placement feedback should be improved.

Insight 3 — Logical hierarchy is clearer than spatial hierarchy

Most participants understood the progression between Topic, Episode, Chapter and Podcast Experience.

However, uncertainty occurred at some transitions, particularly between Chapter selection and the Current Chapter Door. One participant also described room-to-room teleportation as abrupt and suggested a spatial map.

Therefore, users generally understand **what level they are navigating**, but not always **where spaces are in relation to each other**. Stronger current-location and transition cues are needed.

Insight 4 — The concept is spatial, but parts of the experience remain too 2D/static

Participants responded positively to embodied interactions such as doors, while conventional panels and limited environmental movement were perceived as less immersive. Some participants specifically wanted greater movement and richer environmental content.

The next iteration should therefore prioritise **physical and spatial interaction**, while using flat UI only where it clearly supports usability.

4. Evaluation of Aims

The overall testing objective was **partially validated**. First-time users generally understood the XR podcast concept and completed the main exploration flow, but several interactions still required assistance.

Testing Aim	Evaluation	Evidence
Understand Topic Doors	Validated	Door entry was consistently understandable and positively received.
Browse/select Episodes	Validated	Episode browsing and selection were generally completed successfully.
Understand Chapter navigation	Partially Validated	Chapter meaning and the transition to the Current Chapter Door caused uncertainty.
Independently complete Favourite	Invalidated	Multiple participants required assistance with the Heart interaction.
Understand return navigation	Partially Validated	Purpose was understood, but panel behaviour and spatial context caused difficulties.

Testing Aim	Evaluation	Evidence
Understand spatial hierarchy	Partially Validated	Logical flow was clear, but some transitions felt abrupt or insufficiently spatial.

Overall, testing **validated the core spatial podcast concept and door metaphor**, while identifying Favourite affordance, Chapter transition cues and spatial continuity as the main areas requiring iteration.

5. Concept Iteration

Evaluation findings were translated into design changes according to their impact on independent task completion.

Insight / Problem	Design Response	Priority
Heart action is unclear	Make Heart larger and clearly 3D ; improve grab affordance and feedback	High
Favourite result is unclear	Keep the placed Heart visibly attached to the Favourite Area	High
Chapter → Door transition is unclear	Highlight Current Chapter Door after selection and add an “Enter selected chapter” cue	High
Spatial relationships are weak	Explore a spatial/map-based Navigation representation	Medium
Experience feels static	Increase opportunities for spatial movement and environmental exploration	Medium
Supporting UI needs refinement	Improve Navigation layout, Episode presentation and subtitle controls	Low–Medium

The highest-priority changes directly address interactions that prevented independent task completion.

Environmental richness and supporting UI improvements remain important but will follow the core usability fixes.

The successful **door-based interaction will be retained**, demonstrating that evaluation does not only identify features to remove or fix; it also provides evidence for design decisions that should be affirmed.

6. Reflection and Next Steps

This evaluation demonstrated that **task completion alone does not indicate successful usability**. Some participants eventually completed tasks but required hesitation, clarification or assistance. Observing how users approached each interaction was therefore more informative than recording success alone.

The testing also revealed a difference between designing an interaction that appears understandable during development and making it discoverable to a first-time user. The Heart was the clearest example: its meaning was recognisable, but its physical action was not.

A limitation of this round was that some sessions were affected by prototype setup/state issues, including Chapter labelling and Navigation availability. A consistent **pre-test reset checklist** will be used in the next evaluation so that design issues can be more clearly separated from implementation problems.

The next prototype will prioritise **Heart affordance, Chapter-to-door transition cues, navigation clarity and spatial movement**. Evaluation 2 will test whether these changes reduce hesitation and assistance while preserving the successful door interaction. The goal is to move from a prototype whose overall structure is understandable to an XR experience that users can navigate **independently, spatially and confidently**.

Appendix A: Testing results (raw data)

https://docs.google.com/document/d/195JMWZlxIkJW8yHgaesO_brDy3QKFHqw9bcjfy10aI/edit?usp=sharing